



INVESTIGATING THE UNIPOLAR VERSUS BIPOLAR MEASUREMENT STYLE IN WORD CLASSIFICATION

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Abstract

Emotion in language is typically measured along a single bipolar valence scale that collapses positive and negative affect to opposite poles. This approach treats neutral words as emotionally inert, yet many linguistically “neutral” words may, in fact, evoke simultaneous positive and negative associations, a phenomenon known as ambivalence. This thesis investigates how different operationalizations of affect influence the evaluation of computational word representations. To evaluate word representation models, a supervised regression benchmark was applied to a comprehensive Dutch emotion norms dataset ($N = 23,986$; [Speed & Brysbaert, 2024](#)). Three progressively complex representation families (character n-grams, fastText static embeddings, and BERTje transformer embeddings) were tested on their ability to predict bipolar Valence, unipolar Positive and Negative Affect, alongside a derived Ambivalence score. The results show that fastText embeddings achieve the highest alignment with human ratings for bipolar Valence ($r = .71$), while all models struggle substantially with Ambivalence ($r \leq .27$). Critically, a descriptive analysis reveals that 18.69% of near-neutral words ($\text{Valence} \in [2.75, 3.25]$) exhibit high ambivalence, demonstrating that bipolar neutrality masks genuine emotional complexity. Robustness checks confirm that these findings are sensitive to both the ambivalence formula and the neutral threshold, underscoring the need for transparent measurement choices. This work contributes to an underexplored niche: applying the bipolar–unipolar measurement debate to Dutch-language NLP, a language for which large-scale emotion norm benchmarks have only recently become available.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The dataset utilized in this thesis is the publicly available Dutch word emotion norms dataset published by [Speed and Brysbaert](#)

(2024), accessed via their OSF repository (<https://osf.io/9htuv/>). This dataset contains emotion evaluations for about 24,000 words that were gathered from Dutch-speaking participants. There was no fresh data collected from humans or animals for this thesis. The original owners retain all rights to the data.

Figures and Code. All figures presented in this thesis were generated by the author using Python visualization libraries (matplotlib, seaborn). The complete analysis pipeline, including target construction, embedding extraction, cross-validated benchmarking, and robustness checks, was written by the author.

Technology. During the preparation of this work, the author used an AI programming assistant (Gemini) in order to assist with code structure, pipeline architecture, and writing assistance (paraphrasing, grammar, and spelling). After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the final content of the thesis. No other typesetting or generative tools were used outside of the L^AT_EX typesetting environment.

2 INTRODUCTION

The ability to computationally model emotion in language has emerged as a major challenge in natural language processing (NLP), since emotion is a fundamental element of human cognition and communication. Applications range from public opinion mining and customer feedback analysis (Malik & Bilal, 2024) to mental health monitoring and content moderation. At the heart of these applications lies a deceptively simple question: *how should emotion be measured?*

The dominant framework in affective science is the circumplex model of affect (Russell, 1980), which represents emotional experience along two orthogonal dimensions: valence (pleasant–unpleasant) and arousal (calmness–excitement). In this model, valence is inherently *bipolar*: positivity and negativity are treated as opposite ends of a single continuum, making them mutually exclusive. A phrase with a midpoint rating on the scale is considered to be emotionally neutral.

However, there have been theoretical and empirical disputes to this premise. Watson, Clark, and Tellegen (1988) introduced the Positive and Negative Affect Schedule (PANAS), arguing that positive and negative affects are best understood as two independent *unipolar* dimensions rather than as poles of a single axis. Under this framework, an individual or a word can simultaneously evoke both high positive and high negative affect, a state referred to as *ambivalence* (Vaccaro, Kaplan, & Damasio, 2020). More recently, Souders and Yu (2025) provided evidence that allowing

satisfaction and dissatisfaction to coexist yields richer psychological data than forcing them into a bipolar frame.

This distinction has profound implications for NLP. Words that score close to the middle are regarded as a homogeneous "neutral" class when emotion models are trained and assessed solely on bipolar valence. Yet this class may contain two fundamentally different types of words: those that are genuinely emotionally inert (e.g., *tafel* [table], *voet* [foot]) and those that are emotionally complex, evoking both happiness and distress (e.g., *emotioneel* [emotional], *verleden* [past], *stilte* [silence]). If computational models cannot distinguish between these two cases, they risk systematic misinterpretation of language with emotional implications (Fleisig, Abebe, & Klein, 2023).

Another motivation for this project is the linguistic context. The vast majority of affective computing research has been conducted in English, using resources such as the ANEW norms (Warriner, Kuperman, & Brysbaert, 2013). Dutch has historically been underrepresented. Although Moors et al. (2013) published affective norms for 4,300 Dutch words covering valence, arousal, and dominance, these norms did not include discrete emotion categories (happiness, anger, fear, sadness, disgust). The recent publication of the Speed and Brysbaert dataset (Speed & Brysbaert, 2024), covering approximately 24,000 Dutch words with both dimensional and discrete emotion ratings, has opened a unique window for Dutch-language affective benchmarking. This makes Dutch an ideal and still underexplored testing ground for the bipolar–unipolar measurement debate.

Against this background, the present study is designed as a direct computational follow-up to the psychological data collection by Speed and Brysbaert (2024). In their discussion, Speed and Brysbaert explicitly highlighted a key shortcoming of traditional bipolar measurement: it compresses concepts of mixed valence (e.g., “bitter-sweet” feelings, or valence ambiguity; Mattek, Woldford, and Kelley 2017), incorrectly mapping them to the neutral midpoint. They called for future research to explore word processing using different measures of valence, including unipolar measures and measures of valence ambiguity. This thesis directly answers that call. It develops a reproducible human–model alignment benchmark that directly compares bipolar and unipolar-style affect targets. Three word representation families of increasing complexity—character n-grams, fastText static embeddings (Bojanowski, Grave, Joulin, & Mikolov, 2017), and BERTje transformer embeddings (de Vries et al., 2019)—are evaluated using Ridge regression under identical cross-validation conditions. This design ensures that observed performance differences can be attributed to the representations and target operationalisations rather than to differences in model architecture.

Three research questions serve as the study's focus:

- RQ1 *Bipolar vs. unipolar-style*: How does model performance and model ranking change when predicting emotion using bipolar valence versus a unipolar-style affect decomposition (separate positive and negative affect indices)?
- RQ2 *Ambivalence as a test case*: Which model representations best predict ambivalence (mixed affect), and to what extent does ambivalence occur among words that are near-neutral on bipolar valence?
- RQ3 *Robustness*: How sensitive are the findings to specific operational choices, such as the threshold used to define near-neutrality and the mathematical formulation of ambivalence?

3 RELATED WORK

3.1 Dimensional Models of Emotion

For over four decades, the circumplex model of affect (Russell, 1980) has dominated the field of emotion research. It posits that all affective states can be located in a two-dimensional space defined by valence and arousal. Valence, in particular, is conceptualized as a single bipolar dimension in which pleasantness and unpleasantness are perfect opposites. While the circumplex model has been enormously influential, it is not the only way to conceptualize the structure of affect. Scherer (2005) distinguished between discrete emotion theories (which posit a small set of basic emotions such as anger, fear, and happiness) and dimensional theories (which locate emotions in a continuous space). The circumplex model falls into the latter camp, but its bipolar treatment of valence is a specific architectural choice, not a theoretical necessity.

This bipolar assumption was challenged by Watson et al. (1988), who developed the PANAS framework. Positive Affect (PA) and Negative Affect (NA) are regarded as separate, mostly uncorrelated dimensions in PANAS. This means that a person or a stimulus can score high on both PA and NA simultaneously, capturing states of mixed affect that are invisible under bipolar measurement. The theoretical foundation for this independence was formalized by Cacioppo and Berntson (1994) in their *evaluative space model*, they argue that the brain processes positive and negative evaluations through separate neural systems. This approach states that negativity and positivity are not mutually activated; instead, they can be independently engaged, producing states of co-activation (both high), co-inhibition (both low), or the more commonly assumed reciprocal activation (one high, one

low). More recently, [Souders and Yu \(2025\)](#) extended this argument to applied settings, demonstrating that treating satisfaction and dissatisfaction as independent scales captures meaningful variance that bipolar measures miss.

Empirical evidence for the co-occurrence of positive and negative emotions was provided by [Larsen, McGraw, and Cacioppo \(2001\)](#), who demonstrated that people frequently report feeling happy and sad simultaneously, particularly in response to bittersweet life events such as graduating or moving away from home. This finding directly contradicts the bipolar assumption that positive and negative affect are mutually exclusive, and it has implications for how “neutral” words should be interpreted in emotion norm datasets.

The neuroscientific underpinnings of ambivalence have been explored by [Vaccaro et al. \(2020\)](#), who provided neural evidence that the brain can simultaneously process appetitive (approach) and aversive (avoidance) signals. Using functional neuro-imaging, they showed that bittersweet stimuli activate both reward-related and aversion-related brain regions simultaneously, rather than producing a blended “neutral” activation. This challenges the assumption that a “neutral” stimulus is necessarily emotionally inert; instead, it may reflect a balanced co-activation of opposing emotional systems.

3.2 *Affective Norms and the Dutch Language Niche*

Affective norm datasets provide standardized human ratings of how words “feel” along emotional dimensions. The most widely used English resource is the extended ANEW dataset by [Warriner et al. \(2013\)](#), which provides valence, arousal, and dominance ratings for 13,915 English lemmas. [Mohammad \(2018\)](#) subsequently obtained reliable ratings for 20,000 English words, demonstrating that crowd sourced annotations can achieve high inter-annotator reliability when best-worst scaling methods are employed. For Dutch, the first major resource was the [Moors et al. \(2013\)](#) dataset, which provided affective norms for 4,300 Dutch words on valence, arousal, and dominance using 7-point Likert scales.

The publication of the [Speed and Brysbaert \(2024\)](#) dataset represented a significant step forward for Dutch affective research. Covering approximately 24,000 words, this dataset includes not only bipolar Valence and Arousal ratings but also discrete emotion ratings for Happiness, Anger, Fear, Sadness, Disgust, and Surprise. The inclusion of discrete categories is what makes the present study possible: it provides the raw material for constructing independent Positive Affect and Negative Affect variables, enabling a direct test of the bipolar–unipolar distinction.

Despite these resources, Dutch remains a significantly underresourced language in affective NLP compared to English. The EmotioNL dataset (De Bruyne, De Clercq, & Hoste, 2021), consisting of 2,000 annotated tweets and TV captions, is one of the few benchmarks specifically designed for Dutch emotion detection. Research using Dutch-specific transformer models such as BERTje has predominantly focused on tasks like named entity recognition and sentiment classification (de Vries et al., 2019), rather than on the finer-grained question of affect measurement validity. The present study addresses this gap by bringing the bipolar–unipolar measurement debate into the Dutch NLP landscape.

3.3 *Emotion Lexicons and Crowdsourced Norms*

Beyond dimensional norm datasets, several large-scale emotion lexicons have been developed that associate words with categorical emotion labels. The NRC Word–Emotion Association Lexicon (Mohammad & Turney, 2013) is the most widely used, covering over 14,000 English words annotated via crowdsourcing for eight basic emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, trust) as well as positive and negative sentiment. Crucially, the NRC lexicon allows a word to be associated with *multiple* emotions simultaneously, implicitly acknowledging that emotional meaning is not mutually exclusive. However, the NRC lexicon uses binary labels rather than continuous ratings, limiting its ability to capture gradations of affect.

At the sentence level, Buechel and Hahn (2017) introduced EmoBank, a corpus of 10,000 English sentences annotated for valence, arousal, and dominance from both writer and reader perspectives. Their work highlighted that the *perspective* of annotation (who is feeling the emotion) significantly affects the resulting ratings, a nuance that is relevant to word-level norms as well. The Speed and Brysbaert dataset used in the present study reflects the *reader* perspective: participants rated how each word made *them* feel.

These resources share a common limitation: they are overwhelmingly English-language. While the NRC lexicon has been machine-translated into over 100 languages, translation-based affect resources are known to introduce systematic biases, as emotional connotations are often culture- and language-specific (Mohammad, 2018). This underscores the value of natively collected Dutch norms such as those provided by Speed and Brysbaert (2024).

3.4 Word Representations and Affect Prediction

The ability of models to capture affective meaning is directly impacted by the choice of lexical representation. Three broad families can be distinguished:

Form-based representations such as character n-grams capture patterns in how words are spelled. While they cannot encode semantic meaning directly, Dutch morphology contains systematic cues: diminutives (e.g., *-tje*, *-je*) often carry positive connotations, while certain prefixes (e.g., *on-*, *mis-*) are associated with negativity. Term Frequency-Inverse Document Frequency (TF-IDF) character n-gram features have proven surprisingly competitive in text classification tasks (Bojanowski et al., 2017).

Static embeddings represent each word as a fixed dense vector learned from distributional co-occurrence patterns in large corpora. The foundational work on distributed word representations by Mikolov, Sutskever, Chen, Corrado, and Dean (2013) (Word2Vec) demonstrated that such vectors capture meaningful semantic regularities, including analogical reasoning. Pennington, Socher, and Manning (2014) extended this approach with GloVe, which combines global co-occurrence statistics with local context window methods. FastText (Bojanowski et al., 2017) further improved upon these approaches by incorporating subword information: each word vector is computed as the sum of its constituent character n-gram vectors. This subword approach is particularly advantageous for morphologically rich languages such as Dutch, as it can generate vectors for out-of-vocabulary words by composing their character n-gram representations.

Contextual embeddings, such as those produced by BERT (Devlin, Chang, Lee, & Toutanova, 2019) and its language-specific variants, generate context-dependent representations that capture polysemy and syntactic structure. For Dutch, two major pre-trained models are available: BERTje (de Vries et al., 2019), a monolingual Dutch BERT model trained on a diverse Dutch corpus of 2.4 billion tokens, and RobBERT (Delobelle, Winters, & Berendt, 2020), a Dutch RoBERTa model trained on the OSCAR corpus. When applied to isolated words (as in emotion norm prediction), the contextual advantage of these models is reduced, since the representations lack the sentential context that drives their performance gains over static embeddings. Plisiecki and Sobieszek (2024) demonstrated that transformer-based models can extrapolate affective norms with high accuracy, though their benchmarks used bipolar targets only. The present study uses BERTje rather than RobBERT because BERTje was trained on a multi-source Dutch corpus, whereas RobBERT was trained on the Dutch OSCAR corpus; additionally, BERTje is a well-established model in Dutch NLP research (Delobelle et al., 2020; de Vries et al., 2019).

3.5 *Measurement Validity and Annotator Disagreement*

A critical thread connecting these strands of research is the question of *measurement validity*. [Flake and Fried \(2020\)](#) argued that “measurement schmeasurement” is a prevalent problem in psychology: researchers often adopt operationalizations without questioning whether they faithfully capture the theoretical construct of interest. This concern applies directly to NLP, where bipolar valence is routinely used as the sole affect target without considering whether it conflates distinct emotional phenomena.

The NLP community has begun to recognize related issues in the context of annotator disagreement. [Fleisig et al. \(2023\)](#) demonstrated that disagreement in subjective annotations often reflects genuine ambiguity rather than noise. In the context of emotion norms, the standard deviation of ratings is particularly informative for mid-range words, precisely the words where ambivalence is most likely to occur ([Speed & Brysbaert, 2024](#)). By explicitly modeling ambivalence, the present study attempts to honor this disagreement rather than averaging it away.

4 METHODS

The study is designed as a supervised regression benchmark in which computational representations of Dutch words are used to predict human emotion ratings.

4.1 *Dataset*

The primary dataset is the Dutch word emotion norms published by [Speed and Brysbaert \(2024\)](#), comprising $N = 23,986$ unique Dutch words after preprocessing. The dataset includes continuous ratings on a 1–5 scale for bipolar Valence ($M = 2.91$, $SD = 0.64$) and Arousal ($M = 3.03$, $SD = 0.46$), as well as discrete emotion ratings for Happiness ($M = 1.58$, $SD = 0.69$), Anger ($M = 1.39$, $SD = 0.60$), Fear ($M = 1.43$, $SD = 0.60$), Sadness ($M = 1.38$, $SD = 0.57$), Disgust ($M = 1.30$, $SD = 0.48$), and Surprise ($M = 1.37$, $SD = 0.39$). The dataset also includes psycholinguistic control variables such as word frequency (Zipf scale), concreteness, imageability, age of acquisition, and part of speech.

4.2 *Target Construction and Operationalization*

To enable a direct comparison between measurement styles, affect was operationalized in two ways:

Bipolar target. The original Valence rating served as the primary bipolar target. On the 1–5 scale, values below 3.0 indicate negative affect and values above 3.0 indicate positive affect.

Unipolar-style targets. Following the logic of the PANAS framework (Watson et al., 1988), Positive Affect (PA) and Negative Affect (NA) were constructed as independent variables:

$$\text{PositiveAffect} = \text{Happiness} \quad (1)$$

$$\text{NegativeAffect} = \frac{\text{Anger} + \text{Fear} + \text{Sadness} + \text{Disgust}}{4} \quad (2)$$

Using Happiness as the sole proxy for Positive Affect is a pragmatic simplification: the Speed and Brysbaert (2024) dataset does not include other positive discrete emotions (e.g., love, pride). Surprise was excluded from Negative Affect because it does not have a consistently negative valence polarity (Russell, 1980; Speed & Brysbaert, 2024). A sensitivity analysis including Surprise is reported in Section 5.4.

Ambivalence. To quantify the degree of mixed affect, the PA and NA scores were first rescaled to $[0, 1]$ using min-max normalization. The primary ambivalence metric was computed as:

$$\text{Ambivalence}_{\text{product}} = \text{PA}_{01} \times \text{NA}_{01} \quad (3)$$

This product formula yields high values only when both components are simultaneously elevated, capturing the intuition that ambivalence requires the co-occurrence of positive and negative affect (Vaccaro et al., 2020). The product formula is conceptually aligned with the evaluative space model (Cacioppo & Berntson, 1994), which suggests that feeling positive and negative emotions at the same time creates a distinct emotional state. A secondary metric, $\text{Ambivalence}_{\text{min}} = \min(\text{PA}_{01}, \text{NA}_{01})$, was computed for robustness (see Section 5.4).

Construct validity was assessed through correlation analysis (see Appendix A, Table 5 for the full correlation matrix). PA correlated strongly with Valence ($r = .70$) and NA correlated strongly and negatively with Valence ($r = -.76$), confirming expected directionality. Crucially, $\text{Ambivalence}_{\text{product}}$ correlated only weakly with Valence ($r = .18$), confirming that it captures a distinct construct not reducible to bipolar valence.

4.3 Near-Neutral Words

A central question is whether bipolar neutrality masks ambivalence. To investigate this, a subset of near-neutral words was defined using a primary threshold of Valence $\in [2.75, 3.25]$, centered on the scale midpoint of 3.0.

This threshold yielded 10,436 words (43.51% of the vocabulary). Sensitivity analyses were conducted with a narrow threshold ($[2.90, 3.10]$; $n = 4,826$, 20.12%) and a wide threshold ($[2.50, 3.50]$; $n = 15,133$, 63.09%).

4.4 Word Representations

Three representation families were selected, spanning sublexical, distributional, and contextual levels of linguistic information. These models were chosen to provide a progression of complexity, allowing us to test whether advanced neural architectures are necessary for predicting ambivalence, or if simple form-based or distributional models suffice. The specific configurations are as follows:

1. **Character n-grams (form-based baseline).** Words were represented using TF-IDF character n-gram vectors (analyzer = `char_wb`, $n \in [2, 4]$, `min_df = 3`, `max_features = 50,000`). The `char_wb` analyzer was specifically chosen because it extracts n-grams only within word boundaries. This is ideal for single-word evaluation as it effectively captures morphological features like affective prefixes (e.g., *on-*) and suffixes (e.g., *-heid*) without generating boundary-spanning artifacts. The range $n \in [2, 4]$ is standard for capturing these Dutch morphemes, while the frequency thresholds (`min_df = 3`, `max_features = 50,000`) were selected to filter out rare typos while maintaining a rich feature space. This resulted in 17,052 features with 100% vocabulary coverage. Importantly, the TF-IDF vocabulary was fit inside each training fold only, preventing information leakage.
2. **fastText static embeddings.** Pre-trained 300-dimensional Dutch fastText vectors (`cc.nl.300.vec`; [Bojanowski et al., 2017](#)) were used to represent each word as a fixed dense vector. This specific model was chosen as it is the official, widely adopted baseline trained by Facebook AI on massive Dutch corpora (Common Crawl and Wikipedia), ensuring high quality and reproducibility. FastText’s subword decomposition allows it to generate vectors for out-of-vocabulary words by summing constituent character n-gram embeddings. Embeddings were extracted once and cached as an NPZ file.
3. **BERTje transformer embeddings.** Contextual embeddings were obtained from the GroNLP/bert-base-dutch-cased model ([de Vries et al., 2019](#)), a monolingual Dutch BERT model trained on 2.4 billion tokens. A monolingual model was explicitly chosen over multilingual BERT (mBERT) to ensure maximum sensitivity to Dutch linguistic nuances. Each word was passed through the model in isolation, and

the final hidden layer activations were mean-pooled across all sub-tokens to produce a single 768-dimensional vector. Mean-pooling the final hidden layer was chosen because it is a standard, robust method for deriving fixed-size vector representations from transformer architectures when task-specific fine-tuning is not applied. No fine-tuning was performed; the model was used purely for inference-time feature extraction.

4.5 *Modeling Approach*

All representation–target combinations were evaluated under a common supervised regression framework using Ridge regression. The deliberate choice of a single, linear model ensures that performance differences across conditions can be attributed to the quality of the representations and the nature of the target variable, rather than to architectural differences. A DummyRegressor (mean predictor) was included as a lower-bound baseline.

4.6 *Evaluation Protocol*

Model performance was assessed using 5-fold cross-validation with a fixed random seed (42). The same fold indices were reused across all targets and representations to ensure fair comparison. Hyperparameter selection (Ridge $\alpha \in \{0.01, 0.1, 1, 10, 100, 1000\}$) was performed via nested 3-fold cross-validation within each outer training fold, preventing data leakage.

Three complementary metrics were reported: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Pearson correlation (r). MAE and RMSE quantify prediction error in the units of the target variable, while Pearson r captures the linear alignment between predicted and observed values.

5 RESULTS

5.1 *Exploratory Data Analysis*

Before presenting the benchmark results, it is instructive to examine the distributional properties of the target variables, as these directly affect model performance and interpretation.

Figure 1 displays the distributions of bipolar Valence alongside the unipolar-style Positive Affect and Negative Affect targets. Valence follows an approximately symmetric distribution centered near the scale midpoint

($M = 2.91$, $SD = 0.64$, skewness = -0.33), with a slight leftward skew indicating a mild negativity bias in the Dutch lexicon. In contrast, both PA and NA are strongly right-skewed (PA: skewness = 1.73 ; NA: skewness = 2.06), with the majority of words scoring near the floor of the scale. This reflects the fact that most words in a language are emotionally neutral or mildly affective, with only a small subset carrying strong emotional content.

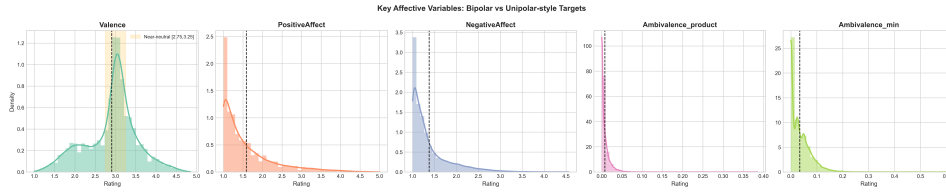


Figure 1: Distribution of bipolar Valence versus unipolar-style Positive Affect and Negative Affect. Valence is approximately symmetric, while PA and NA are strongly right-skewed.

The $\text{Ambivalence}_{\text{product}}$ variable is extremely skewed ($M = 0.0075$, $SD = 0.012$, skewness = 5.93 , kurtosis = 98.52), with the vast majority of words scoring at or near zero. This distributional property has important implications for regression modeling: it means that the ambivalence target is dominated by near-zero values, making it difficult for any model to improve substantially over a trivial mean predictor in terms of absolute error. The inter-quartile range is only 0.010 , and the 75th percentile is $Q_{75} = 0.0104$.

Figure 2 presents the Pearson correlation matrix among all primary variables. The strong negative correlation between Valence and NA ($r = -.76$) and the strong positive correlation between Valence and PA ($r = .70$) confirm that the constructed unipolar targets are directionally aligned with bipolar valence. However, the weak correlation between $\text{Ambivalence}_{\text{product}}$ and Valence ($r = .18$) confirms that ambivalence captures a distinct construct not reducible to the bipolar dimension (see also Table 5 in Appendix A for the full matrix).

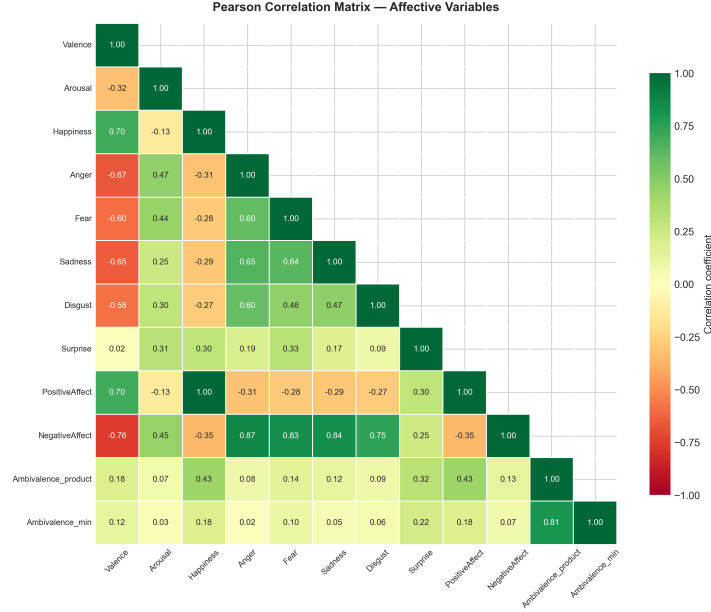


Figure 2: Pearson correlation heatmap among primary affect variables ($N = 23,986$). The weak correlation between Ambivalence and Valence ($r = .18$) confirms construct distinctness.

The dataset includes a wide variety of word classes, including 59.87% nouns, 19.76% verbs, 17.55% adjectives, 2.58% function words, and 0.24% numerals. This broad coverage ensures that the benchmark results are not driven by a single grammatical category.

5.2 Bipolar versus Unipolar Prediction (RQ1)

Table 1 presents the full benchmark results across all target–representation combinations. Across all targets, fastText static embeddings and BERTje transformer embeddings substantially outperformed both the character n-gram baseline and the dummy mean predictor.

Table 1: Benchmark results: mean (SD) across 5-fold cross-validation. Best Pearson r per target is in **bold**.

Target	Representation	MAE	RMSE	Pearson r
Valence	Dummy (mean)	0.486	0.639	0.000
	Char n-grams	0.403	0.532	0.555
	fastText	0.344	0.453	0.708
	BERTje	0.376	0.493	0.637
PositiveAffect	Dummy (mean)	0.517	0.689	0.000
	Char n-grams	0.446	0.602	0.486
	fastText	0.396	0.539	0.627
	BERTje	0.426	0.576	0.549
NegativeAffect	Dummy (mean)	0.338	0.465	0.000
	Char n-grams	0.286	0.398	0.530
	fastText	0.244	0.341	0.682
	BERTje	0.262	0.364	0.623
Ambivalence _{prod}	Dummy (mean)	0.008	0.012	0.000
	Char n-grams	0.008	0.012	0.246
	fastText	0.007	0.012	0.270
	BERTje	0.008	0.012	0.239

For **bipolar Valence**, fastText achieved the highest correlation ($r = .708$), followed by BERTje ($r = .637$) and character n-grams ($r = .555$). All three representations substantially outperformed the dummy baseline. The model ranking (fastText > BERTje > n-grams) was consistent across targets.

For the **unipolar-style targets**, NegativeAffect was predicted more accurately than PositiveAffect across all representations. For instance, fastText achieved $r = .682$ for NA versus $r = .627$ for PA. This asymmetry may reflect the broader distributional spread of negative emotions across four component variables (Anger, Fear, Sadness, Disgust), compared to the single-variable proxy (Happiness) used for PA.

Ambivalence was the most difficult target to predict. Even the best-performing model (fastText, $r = .270$) achieved only a modest correlation. The MAE and RMSE values for ambivalence were very low in absolute terms ($\text{MAE} \approx 0.008$), reflecting the highly skewed distribution of the variable (see Section 5.1). The near-zero baseline MAE means that a trivial mean predictor is almost as accurate as the trained models in absolute error, even though the trained models do capture some meaningful rank-order signal.

5.3 Ambivalence and Near-Neutrality (RQ2)

The descriptive analysis of near-neutral words revealed a critical insight. Using the primary threshold (Valence $\in [2.75, 3.25]$), 10,436 words (43.51% of the full vocabulary) were classified as near-neutral on the bipolar scale. Within this near-neutral subset:

- The mean Positive Affect was 1.44 and the mean Negative Affect was 1.18, both mildly elevated above the floor of 1.0.
- 18.69% of near-neutral words exhibited high ambivalence (above the 75th percentile of the $\text{Ambivalence}_{\text{product}}$ distribution, $Q_{75} = 0.0104$).
- Using the alternative $\text{Ambivalence}_{\text{min}}$ formula, this proportion increased to 27.6%, indicating that the choice of formula meaningfully affects the estimated prevalence of ambivalence.

This finding demonstrates that nearly one in five words classified as “neutral” on a bipolar valence scale actually evokes substantial mixed affect, consistent with the theoretical predictions of [Cacioppo and Berntson \(1994\)](#) and the empirical findings of [Larsen et al. \(2001\)](#). Table 2 illustrates this distinction with concrete Dutch word examples.

Table 2: Example near-neutral words illustrating the contrast between genuinely neutral and ambivalent items. PA = Positive Affect; NA = Negative Affect; Amb = $\text{Ambivalence}_{\text{product}}$.

Word	Translation	Valence	PA	NA	Amb
<i>Near-neutral, Low Ambivalence (genuinely neutral)</i>					
tafel	table	3.05	1.36	1.00	0.000
voet	foot	3.00	1.00	1.00	0.000
boom	tree	3.16	1.95	1.00	0.000
vier	four	3.05	1.63	1.00	0.000
<i>Near-neutral, High Ambivalence (emotionally complex)</i>					
emotioneel	emotional	2.95	3.33	2.38	0.225
gezichtsuitdrukking	facial expression	3.16	2.45	2.30	0.132
reddingsboot	lifeboat	3.11	2.69	1.87	0.103
stilte	silence	3.05	2.44	1.92	0.093
sentiment	sentiment	3.16	2.40	1.93	0.091
verleden	past	3.00	2.44	1.81	0.082

Words like *emotioneel* (emotional) and *verleden* (past) have bipolar Valence scores very close to 3.0 (midpoint), appearing “neutral.” Yet they show markedly elevated Positive Affect and Negative Affect simultaneously, resulting in high Ambivalence scores. Under a purely bipolar

framework, these words would be indistinguishable from genuinely neutral words like *tafel* (table) or *voet* (foot). Notably, even concrete words like *reddingsboot* (lifeboat) exhibit high ambivalence: the word evokes both hope (rescue) and distress (danger at sea), yielding a near-neutral bipolar score that masks its emotional complexity.

Figure 3 visualizes the relationship between bipolar Valence and $\text{Ambivalence}_{\text{product}}$ across the full vocabulary. The characteristic inverted-U shape confirms that ambivalence peaks near the valence midpoint: words at the extremes of the valence scale tend to be emotionally purely positive or purely negative, while mid-range words are more likely to be ambivalent.

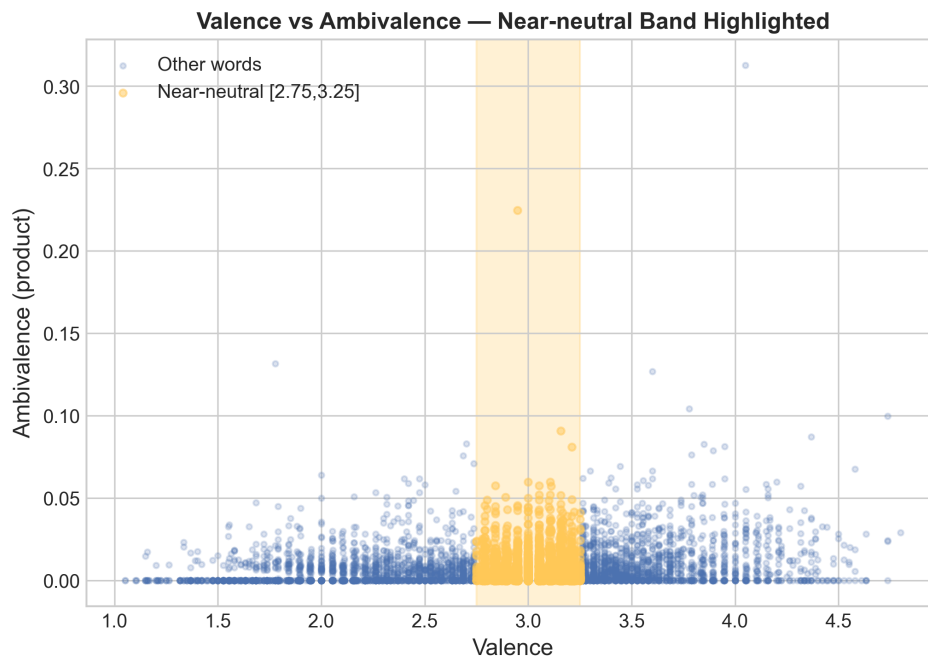


Figure 3: Scatterplot of bipolar Valence versus $\text{Ambivalence}_{\text{product}}$, with near-neutral words highlighted. Ambivalence peaks near the valence midpoint, confirming that bipolar neutrality masks emotional complexity.

Figure 4 further illustrates the independence of Positive and Negative Affect. Under a strict bipolar assumption, all words would fall along the negative diagonal (high PA implies low NA, and vice versa). Instead, the scatterplot reveals substantial scatter, with a meaningful cluster of words exhibiting simultaneously elevated PA and NA—precisely the ambivalent words that bipolar measurement conflates with genuine neutrality.

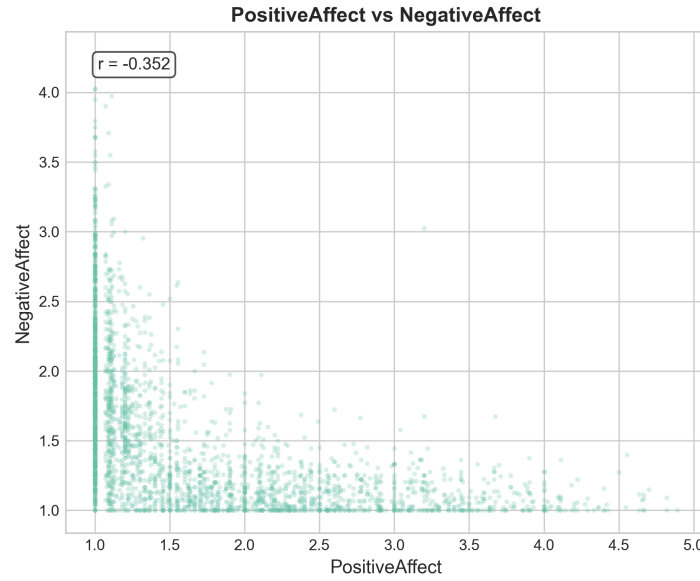


Figure 4: Scatterplot of Positive Affect versus Negative Affect ($N = 23,986$). Under a strict bipolar assumption, all points would fall along the negative diagonal. The scatter and the upper-right quadrant cluster demonstrate the independence of PA and NA.

An analysis of the most ambivalent words in the dataset (Table 3) further illustrates the construct. The top-scoring words are overwhelmingly emotion-related terms (*emoties*, *emotie*, *gevoelens*, *gevoel*) that inherently reference both positive and negative affective experience. This makes semantic sense: words that emphasise emotions should, in fact, elicit both positive and negative connections.

Table 3: Top 10 most ambivalent words in the dataset, ranked by $Ambivalence_{product}$. PA = Positive Affect; NA = Negative Affect.

Word	Translation	Valence	PA	NA	Amb
emoties	emotions	3.47	3.56	3.15	0.386
emotie	emotion	4.05	3.20	3.03	0.313
gevoelens	feelings	3.79	3.22	2.78	0.277
gillen	screaming	2.20	2.82	3.05	0.261
gevoel	feeling	3.74	3.10	2.65	0.243
emoticon	emoticon	3.22	3.85	2.14	0.229
emotioneel	emotional	2.95	3.33	2.38	0.225
gemoedstoestand	state of mind	3.30	2.64	2.23	0.141
gezichtsuitdrukking	facial expression	3.16	2.45	2.30	0.132
regenwoud	rainforest	3.60	3.33	1.81	0.132

5.4 Robustness and Sensitivity (RQ3)

Ambivalence formula comparison. The Pearson correlation between $\text{Ambivalence}_{\text{product}}$ and $\text{Ambivalence}_{\text{min}}$ was $r = .81$ (Spearman $\rho = .92$). While the two formulas are strongly correlated, they diverge meaningfully: the min-based formula is more conservative, yielding higher absolute ambivalence scores and flagging a larger proportion of words as highly ambivalent (27.6% of near-neutral words vs. 18.69% for the product formula). Agreement among the top quartile was 85.95%. Figure 5 visualizes this divergence.

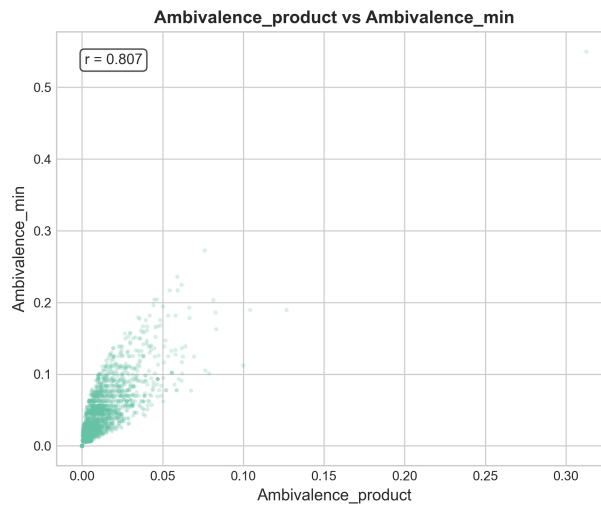


Figure 5: Scatterplot comparing $\text{Ambivalence}_{\text{product}}$ and $\text{Ambivalence}_{\text{min}}$ ($N = 23,986$). Despite a Pearson $r = .81$, the two formulas diverge for extreme values, with the min-based formula producing higher absolute scores.

Neutral threshold sensitivity. The proportion of near-neutral words classified as highly ambivalent varied systematically across threshold widths, as summarized in Table 4. The maximum difference of 6.1 percentage points across thresholds exceeds 5%, indicating meaningful sensitivity to the operational definition of “near-neutrality.”

Table 4: Threshold sensitivity analysis: proportion of near-neutral words classified as highly ambivalent (above Q_{75}) under three threshold definitions.

Threshold	Valence range	N	% of total	% high ambivalence
Narrow	[2.90, 3.10]	4,826	20.12	16.12
Primary	[2.75, 3.25]	10,436	43.51	18.69
Wide	[2.50, 3.50]	15,133	63.09	22.23

The monotonic increase in the proportion of highly ambivalent words as the threshold widens is expected: a broader range captures words further from the midpoint that are more likely to have elevated affect on at least one pole. This pattern underscores the importance of documenting and justifying threshold choices, as recommended by [Flake and Fried \(2020\)](#).

Scaling sensitivity. Ambivalence scores computed under min-max scaling versus z-score scaling (shifted to $[0, 1]$) were perfectly correlated ($r = 1.0$), confirming that the ambivalence construct is robust to the choice of normalization method.

Surprise inclusion. Including Surprise in the Negative Affect composite reduced the correlation with Valence from $r = -.76$ to $r = -.71$, weakening the construct’s discriminant validity. This supports the decision to exclude Surprise from the primary operationalization, consistent with the observation that Surprise is valence-ambiguous: it can accompany both positive events (e.g., a surprise birthday party) and negative events (e.g., an unexpected diagnosis) ([Scherer, 2005](#)).

6 DISCUSSION

The results of this study underscore the profound impact of measurement choices on the evaluation of computational language models. In this section, the findings are interpreted in light of the research questions and the broader literature.

6.1 Representation Quality Across Measurement Styles (RQ1)

One notable result is that fastText consistently outperforms BERTje on all targets. One explanation is the nature of the task: predicting word-level emotion norms from isolated words favors representations that encode stable, context-independent semantic associations. FastText vectors are trained to capture distributional word meaning across large corpora, producing a single, fixed representation per word that directly encodes its typical usage contexts ([Bojanowski et al., 2017](#)). By contrast, earlier static embedding methods such as Word2Vec ([Mikolov et al., 2013](#)) and GloVe ([Pennington et al., 2014](#)) do not incorporate subword information, which may explain why fastText is particularly well-suited to Dutch, where morphological composition (e.g., compound nouns like *reddingsboot* = *redding* + *boot*) carries semantic and affective information.

BERTje, by contrast, is designed to produce *context-dependent* representations. When applied to isolated words without sentential context, its advantage is diminished. This finding aligns with [Plisiecki and Sobieszek \(2024\)](#), who noted that transformer models excel when predicting affect in

context but may not dominate in word-level norm prediction. It is worth noting that an alternative Dutch transformer model, RobBERT (Delobelle et al., 2020), was not evaluated in this study; future work could compare BERTje and RobBERT to determine whether the choice of pre-training corpus or architecture affects prediction performance.

The model ranking was stable across bipolar and unipolar targets, suggesting that the quality of word representations generalizes across measurement styles. However, the absolute performance levels differed: Negative Affect was predicted more accurately than Positive Affect across all models. This asymmetry likely reflects the richer composite operationalization of NA (mean of four emotions) compared to the single-variable proxy for PA (Happiness only), a limitation imposed by the dataset structure (Speed & Brysbaert, 2024).

6.2 The Ambivalence Blind Spot (RQ2)

The most striking finding is the high prevalence of ambivalence among near-neutral words. The observation that 18.69% of words with midpoint bipolar valence actually exhibit high mixed affect empirically confirms the shortcoming identified by Speed and Brysbaert (2024) in their discussion: a bipolar scale compresses valence ambiguity (Mattek et al., 2017). It is also consistent with the evaluative space model’s prediction that co-activation of positive and negative substrates is a common and psychologically meaningful state (Cacioppo & Berntson, 1994), and with concerns raised by Vaccaro et al. (2020) and Souders and Yu (2025). By directly answering the call of Speed and Brysbaert (2024) to contrast unipolar and bipolar measures, this study demonstrates the consequence of this compression in Dutch NLP: when computational models are trained exclusively on bipolar valence, they are structurally incapable of distinguishing genuinely neutral words from emotionally complex ones, forcing both into the same midpoint prediction.

The extremely low predictive correlations for Ambivalence ($r \leq .27$) indicate that current word representations, whether form-based, distributional, or contextual, do not effectively encode the co-activation of opposing emotional valences. This suggests that ambivalence may be a genuinely difficult construct to predict from lexical form alone, potentially requiring compositional or sentential context to resolve. The observation that the top-scoring ambivalent words are predominantly *meta-emotional* terms (e.g., *emoties*, *gevoelens*, *gevoel*) suggests that ambivalence may be more predictable for certain semantic categories than for the vocabulary as a whole.

6.3 *The Dutch Language Niche*

This study contributes to a notably underexplored area. While bipolar–unipolar debates have been conducted primarily in English-language psychology (Souders & Yu, 2025; Watson et al., 1988), and computational affect prediction has been dominated by English resources (Mohammad, 2018; Warriner et al., 2013), the intersection of these two threads in a non-English language is rare. Prior Dutch affective NLP has focused on sentiment classification (De Bruyne et al., 2021) and dimensional norm prediction (Moors et al., 2013), but not on the specific question of whether bipolar measurement obscures ambivalence. The availability of the Speed and Brysbaert (2024) dataset, with its unprecedented combination of dimensional *and* discrete emotion ratings for a large Dutch vocabulary, makes this investigation uniquely feasible.

It is also noteworthy that existing emotion lexicon resources, such as the NRC Word–Emotion Association Lexicon (Mohammad & Turney, 2013), use binary labels for Dutch words that were machine-translated from English. Such translation-based resources may not faithfully capture the emotional connotations of Dutch words, which are shaped by cultural and linguistic context. The natively collected ratings in the Speed and Brysbaert (2024) dataset avoid this issue, providing a more ecologically valid foundation for Dutch affective benchmarking.

6.4 *Implications for Emotion Lexicon Design*

The findings of this study have implications beyond the specific benchmark reported here. The observation that ambivalence is prevalent and poorly predicted suggests that emotion lexicon resources should consider adopting unipolar annotation schemes that allow independent ratings of positive and negative affect, rather than forcing annotators to choose a single point on a bipolar scale. This recommendation aligns with the annotation guidelines of EmoBank (Buechel & Hahn, 2017), which separately annotates writer and reader emotion perspectives, and with the NRC lexicon’s (Mohammad & Turney, 2013) allowance for multiple emotion labels per word.

For practical NLP applications such as opinion mining (Malik & Bilal, 2024), the distinction between genuine neutrality and ambivalence could be consequential. A product review containing ambivalent language (simultaneously positive and negative) should be interpreted differently from a genuinely neutral review, yet current bipolar sentiment classifiers treat both as equivalent. Incorporating unipolar affect dimensions into sentiment analysis pipelines could enable more nuanced and accurate interpretation of user-generated content.

6.5 *Measurement Transparency*

The robustness analyses reveal that ambivalence findings are sensitive to operational choices. The 6.1 percentage-point variation in high-ambivalence rates across threshold definitions and the meaningful divergence between product and min formulas reinforce the argument by [Flake and Fried \(2020\)](#) that measurement decisions require explicit documentation and justification. This study attempts to model this transparency by reporting multiple operationalizations and sensitivity analyses.

6.6 *Limitations*

Several limitations should be acknowledged. First, Positive Affect is operationalized using a single variable (Happiness), which may underrepresent the full breadth of positive emotional experience. Future datasets including additional positive emotions (e.g., love, pride, amusement) would enable a more balanced decomposition. Second, the reliance on Ridge regression, while deliberately chosen to isolate representation quality, limits the ability to capture non-linear interactions between Positive and Negative Affect that may be relevant for ambivalence prediction. Third, BERTje embeddings were extracted for words in isolation, without sentential context, which may underestimate the model's capacity for affect-sensitive representation. Fourth, min-max normalization parameters were fit on the full dataset during target construction; while this does not affect the cross-validated model evaluation (which uses pre-computed targets), it introduces a minor methodological inconsistency.

6.7 *Future Directions*

Future work should investigate whether contextual language models can better disambiguate ambivalence when words are presented in sentence or discourse contexts rather than in isolation. Additionally, non-linear modeling approaches (e.g., gradient-boosted trees, neural networks) may capture interaction effects between PA and NA that linear models miss. Cross-linguistic replication—applying the same bipolar-unipolar benchmark to English, German, or other languages with available discrete emotion norms—would strengthen the generalizability of the findings. Finally, the extremely skewed distribution of the ambivalence target variable suggests that alternative regression frameworks (e.g., zero-inflated models) may be more appropriate for this construct.

7 CONCLUSION

This thesis developed a reproducible benchmark comparing bipolar and unipolar-style affect measurement within Dutch-language computational word classification. The primary goal was to answer three specific research questions regarding model performance, ambivalence, and measurement robustness.

Answering RQ1 (Bipolar vs. unipolar-style prediction): The model ranking was remarkably stable across both bipolar and unipolar targets. Across all dimensions, modern word representations achieved strong alignment with human emotion ratings, with fastText static embeddings consistently outperforming both BERTje transformer embeddings and the character n-gram baseline in this word-level, context-free setting.

Answering RQ2 (Ambivalence as a test case): The unipolar-style decomposition explicitly revealed that 18.69% of words classified as “neutral” on a bipolar valence scale actually exhibit high ambivalence (mixed affect). This explicitly answers RQ2 by demonstrating that bipolar measurement systematically obscures emotional complexity. Furthermore, current NLP representations struggled to predict this ambivalence accurately (all Pearson $r \leq .27$), indicating a critical blind spot in how these models represent emotionally complex vocabulary.

Answering RQ3 (Robustness): The findings are highly sensitive to specific operational choices. The prevalence of ambivalence varied by over 6 percentage points depending on the neutral threshold, and shifting from a product-based to a min-based mathematical formulation increased the high-ambivalence subset from 18.69% to 27.6%. This explicit sensitivity reinforces the need for transparent reporting of measurement decisions in affective NLP.

These findings have practical implications for the development and evaluation of emotion-aware NLP systems. As such systems are increasingly deployed for public opinion analysis, customer feedback interpretation, and content moderation, the ability to distinguish genuine neutrality from emotional ambivalence is not merely an academic concern; it directly affects the validity of automated inferences about human affect. This work demonstrates that moving beyond bipolar valence, even with simple analytical tools, can reveal meaningful emotional structure that would otherwise remain hidden.

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A CORRELATION MATRIX

Table 5 reports the Pearson correlations among all primary variables, confirming construct validity. These correlations are discussed in Sections 4.2 and 5.1.

Table 5: Pearson correlations among primary affect variables ($N = 23,986$).

	Val	PA	NA	Amb _{prod}	Amb _{min}
Valence	1.00	.70	−.76	.18	.12
PositiveAffect	.70	1.00	−.35	.43	.18
NegativeAffect	−.76	−.35	1.00	.13	.07
Amb _{prod}	.18	.43	.13	1.00	.81
Amb _{min}	.12	.18	.07	.81	1.00

The low correlation between Ambivalence_{product} and Valence ($r = .18$) confirms that ambivalence captures variance distinct from bipolar valence, supporting the thesis that a unipolar-style decomposition reveals emotional structure not visible under bipolar measurement.